MILITARY SPECIFICATION SHEET

ELECTRON TUBE, GAS SWITCHING

TYPE 5863

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: TR, band bass, frequency range 8, 490 to 9, 578 MHz, incident power 250 kw

ABSOLUTE RATINGS:

Parameter:	Incident power	Ebb	Alt
Unit:	kw	Vdc	ft
Maximum:	1, 000	---	10, 000
Minimum:	4	-650	---

PHYSICAL CHARACTERISTICS:

Dimensions: See figure 1

TEST CONDITIONS:

Parameter:	Incident power	tp1	Prr	Iz	σ'	F
Unit:	kw	μs	pps	μAdc	---	MHz
Test condition 1:						
Maximum:	44	1.15	---	---	1.10	---
	40	1.00	1, 000	100	---	F7
Minimum:	36	0.85	---	---	---	---
Test condition 2:						
Maximum:	---	1.15	---	---	---	---
	200	1.00	1, 000	100	---	F7
Minimum:	---	0.85	---	---	---	---
Test condition 3:						
Maximum:	1, 000	1.15	---	---	---	---
	---	1.10	1, 000	100	---	F7
Minimum:	250	0.85	---	---	---	---

Frequency		
F	MHz	±
F1	8, 490	0.1%
F2	8, 565	0.1%
F3	8, 650	0.1%
F4	8, 740	0.1%
F5	8, 880	0.1%
F6	8, 920	0.1%
F7	9, 000	0.1%
F8	9, 180	0.1%
F9	9, 280	0.1%
F10	9, 380	0.1%
F11	9, 487	0.1%
F12	9, 578	0.1%

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Qualification inspection							
1031	High-frequency vibration	1	---		---	---	---	---
	Quality conformance inspection, part 1	5	---					
1027	Temperature cycling (nonoperating)	---	---		---	---	---	---
4416	Insertion loss	---	---	Iz = 0; F = F7	Li	---	0.7	dB
4421	Ignitor interaction (insertion loss)	---	---	Iz = 100 μAdc	ΔLi	---	0.3	dB
4406	Ignitor voltage drop	---	---	Iz = 100 μAdc	Eid	200	375	Vdc
4452	Flat leakage power	---	1	tp2 = 0.5 $\pm$0.15 μs	pf	---	30	mW
4452	Spike leakage energy	---	---	Iz = 100 μAdc	Ws	---	0.15	erg
4494	Position of short	2	1		---	---	$\pm$0.007	Inch
4473	Low-level VSWR	---	---	σ' = 1.05 (max); Iz = 100 μAdc;				
				F = F1 and F12	σ	---	1.9	---
				F = F2, F7, and F11	σ	---	1.6	---
	Quality conformance inspection, part 2							
4101	Dielectric material strain	---	---		---	---	---	---
4401	Ignitor ignition time	---	---	Ebb = -700 Vdc; Rl = 5.5 Meg	t	---	5	sec
4471	Recovery time	3, 4	2		t	---	8	μs
4488	Arc loss	---	2	Incident power = 4 kw	La	---	0.8	dB
4473	Low-level VSWR	---	---	σ' = 1.05 (max); Iz = 100 μAdc; F = F2, F4, F5, F6, F8, F9, and F10	σ	---	1.6	---
	Quality conformance inspection, part 3							
4551	Life test	---	3	Group D	t	500	---	hrs
---	Life-test end points:	---	---					
4471	Recovery time	---	2		t	---	18	μs
4452	Flat leakage power	---	1	tp2 = 0.5 $\pm$0.15 μs	pf	---	30	mW
4416	Insertion loss	---	---	F = F7; Iz = 0	Li	---	0.9	dB
4452	Spike leakage energy	---	---	Iz = 100 μAdc	Ws	---	0.2	erg
1027	Temperature cycling life test	---	---	Group B	---	10	---	Cycles

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NOTES:

1. There shall be no evidence of shorts between keep-alive electrodes as indicated by a short indicator during vibration.

2. With a metal plate shorting the line the position of the VSWR minimum shall be determined. The metal plate shall be replaced by the tube and the position of the VSWR minimum of the flat position of the pulse shall be measured and shall be 0.032 inches (0.81 mm) further from the magnetron within the limits specified.

3. The loss of signal in the tube at the specified time after the pulse shall not be greater than 3 dB in excess of the loss at 800 to 1,000 μs after the pulse.

4. The recovery time shall be measured after 30 minutes of operation under the conditions specified.

5. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0 percent, inspection level II.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - MU
Air Force - 11, 17,
DSA - ES

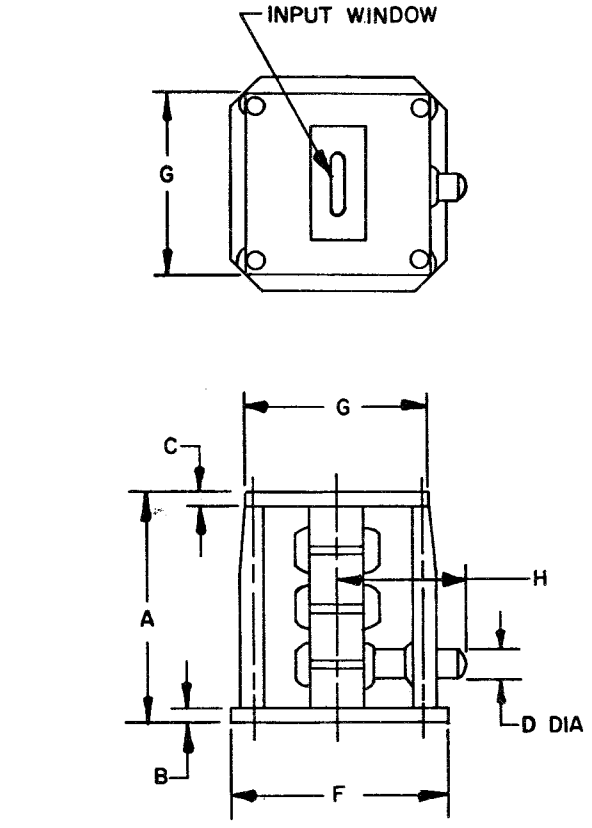
User activities:
Army - WC
Navy - AS, OS, MC, CG, SH

Preparing activity: Navy - EC

Agent: DSA - ES

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Qualification inspection		
B	.100 (2.54)	.120 (3.05)
C	.100 (2.54)	.120 (3.05)
Q	No. 18 Drill .1695 Dia (4.31)	
Quality conformance inspection, part 1		
A	1.966 (49.94)	1.986 (50.44)
Quality conformance inspection, part 2		
H		1.125 (28.58)
J	.735 (18.67)	.739 (18.77)
L	.674 (17.12)	.678 (17.22)
Nominal dimensions		
D	.250 Dia (6.35)	
F	1.875 (47.63)	
G	1.625 (41.28)	
P	1.156 (29.36)	

NOTE:
Exhaust tube shall not extend beyond flange more than .250 (6.35 mm).

FIGURE 1. Outline drawing of electron tube type 5863.

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